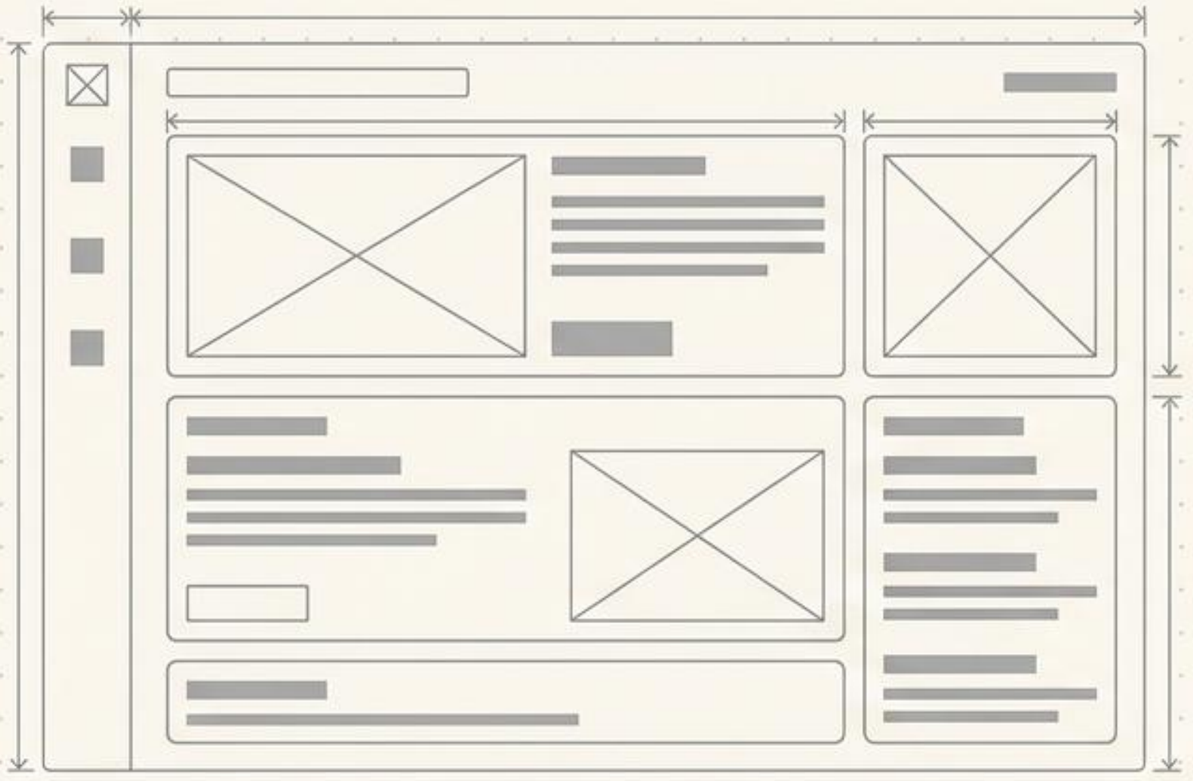




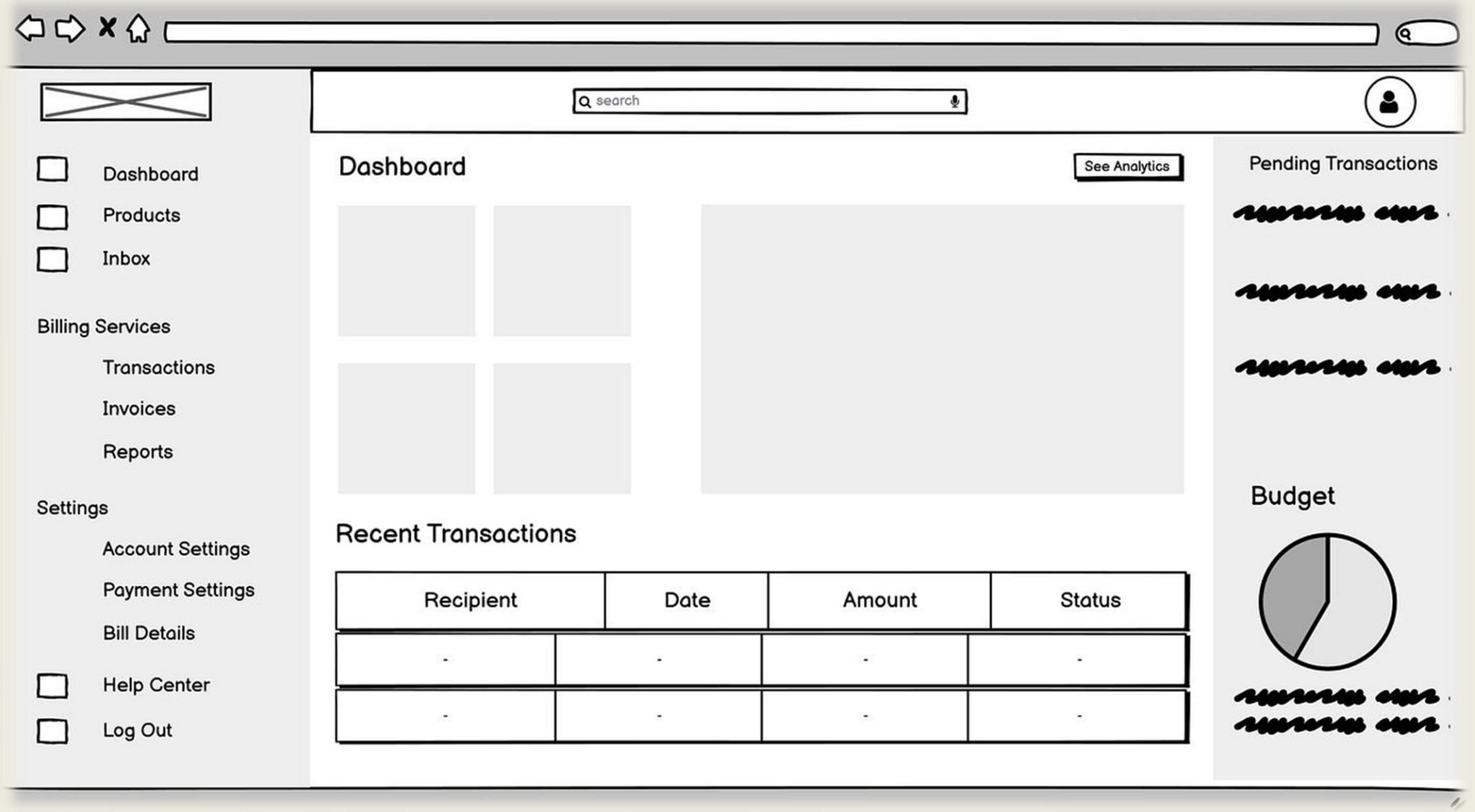
Wireframing with Figma: Blueprinting the Digital World

MIT4301: Turning Ideas into Interfaces (Week 3)

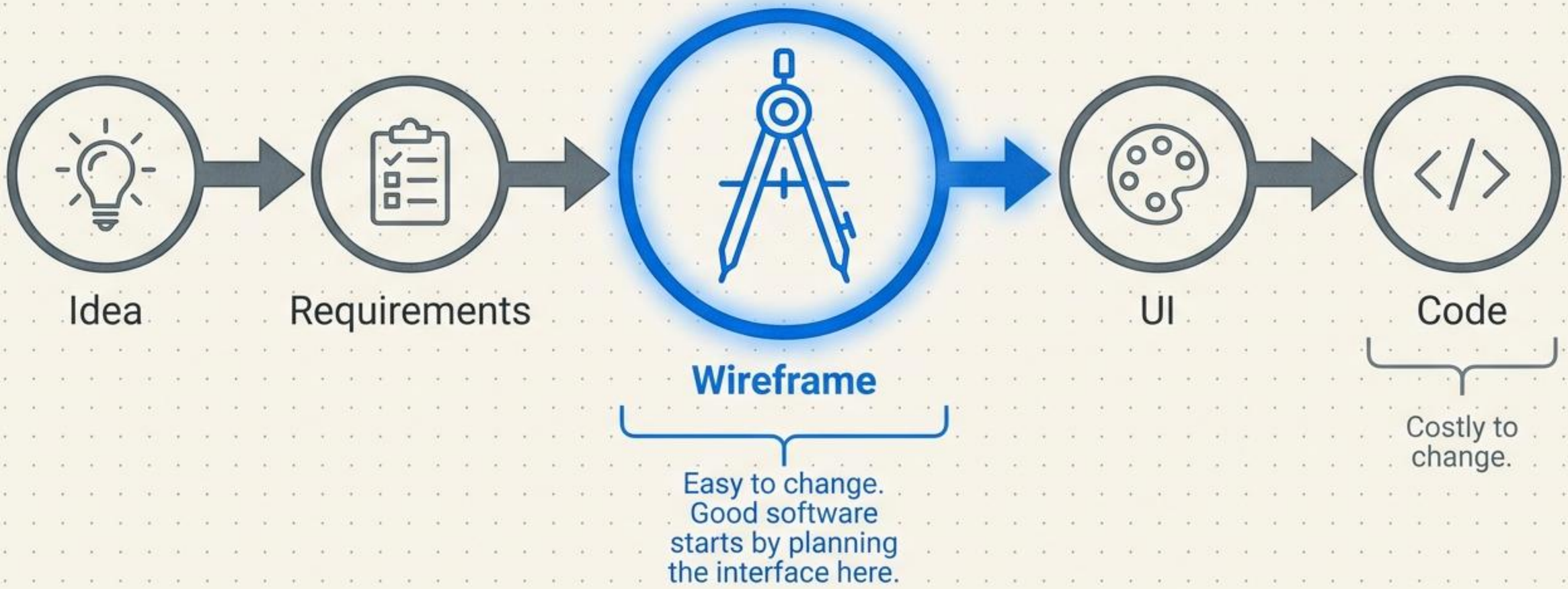
University of Colombo School of Computing

Session Roadmap

What is a wireframe?
Why Figma?
Figma basics
Build a wireframe
Turn it into a UI
Export your design



The Design Pipeline: Plan Before You Pour Concrete

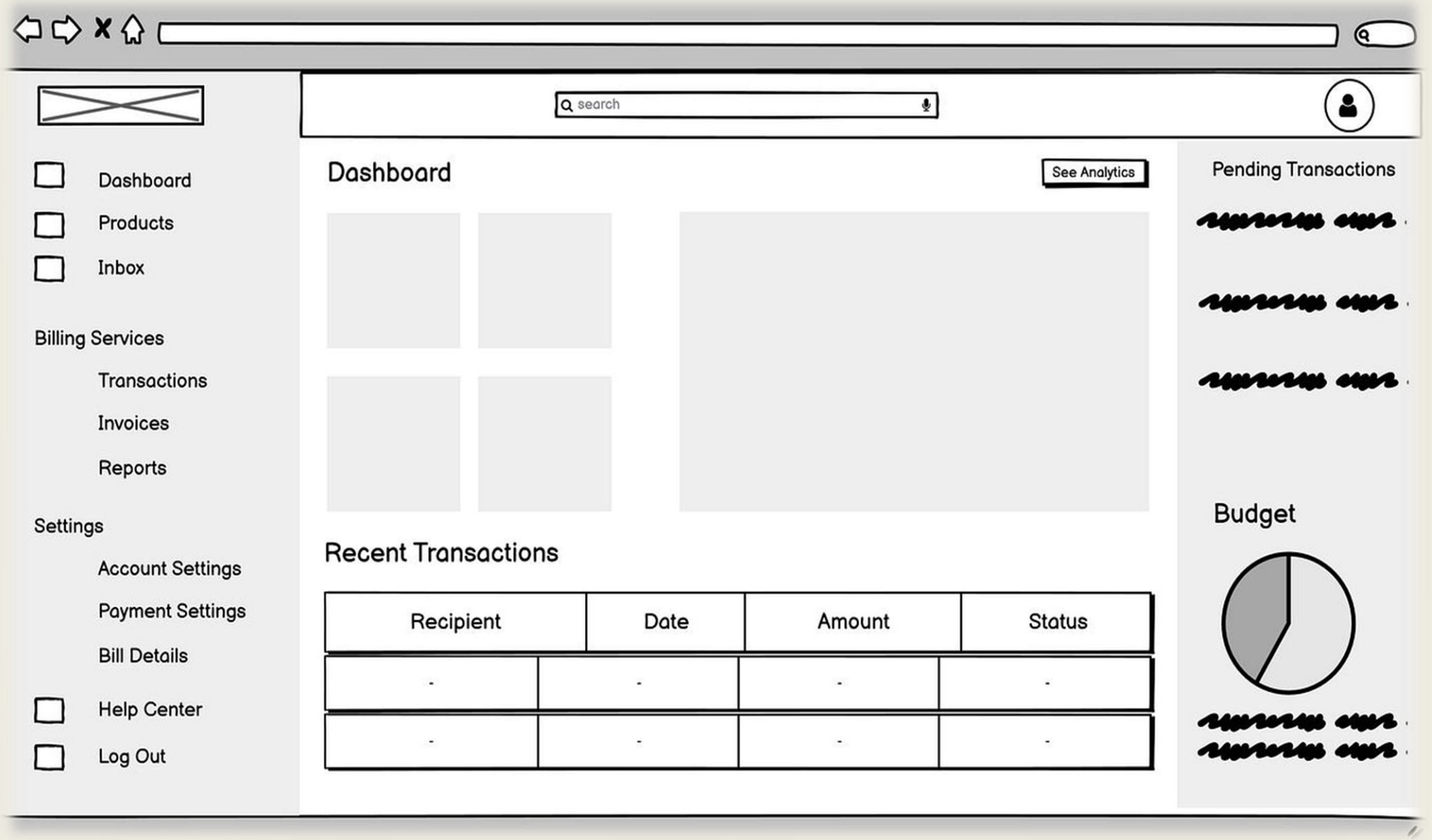


What is a Wireframe?

A visual blueprint of a software interface.

Focus on layout, navigation, buttons, and forms.

No colors or decoration yet.



WIREFRAME

Attribute: Structure



- **Focus:** Layout, navigation, buttons, forms.
- **Constraint:** No colours or decoration.

MOCKUP

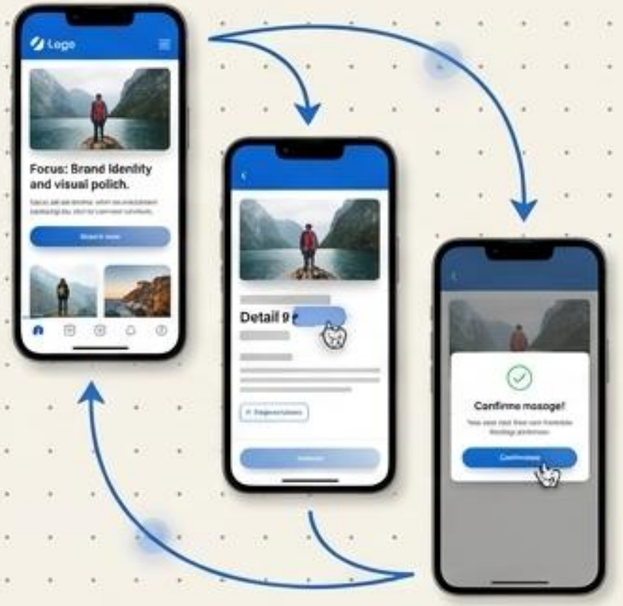
Attribute: Appearance



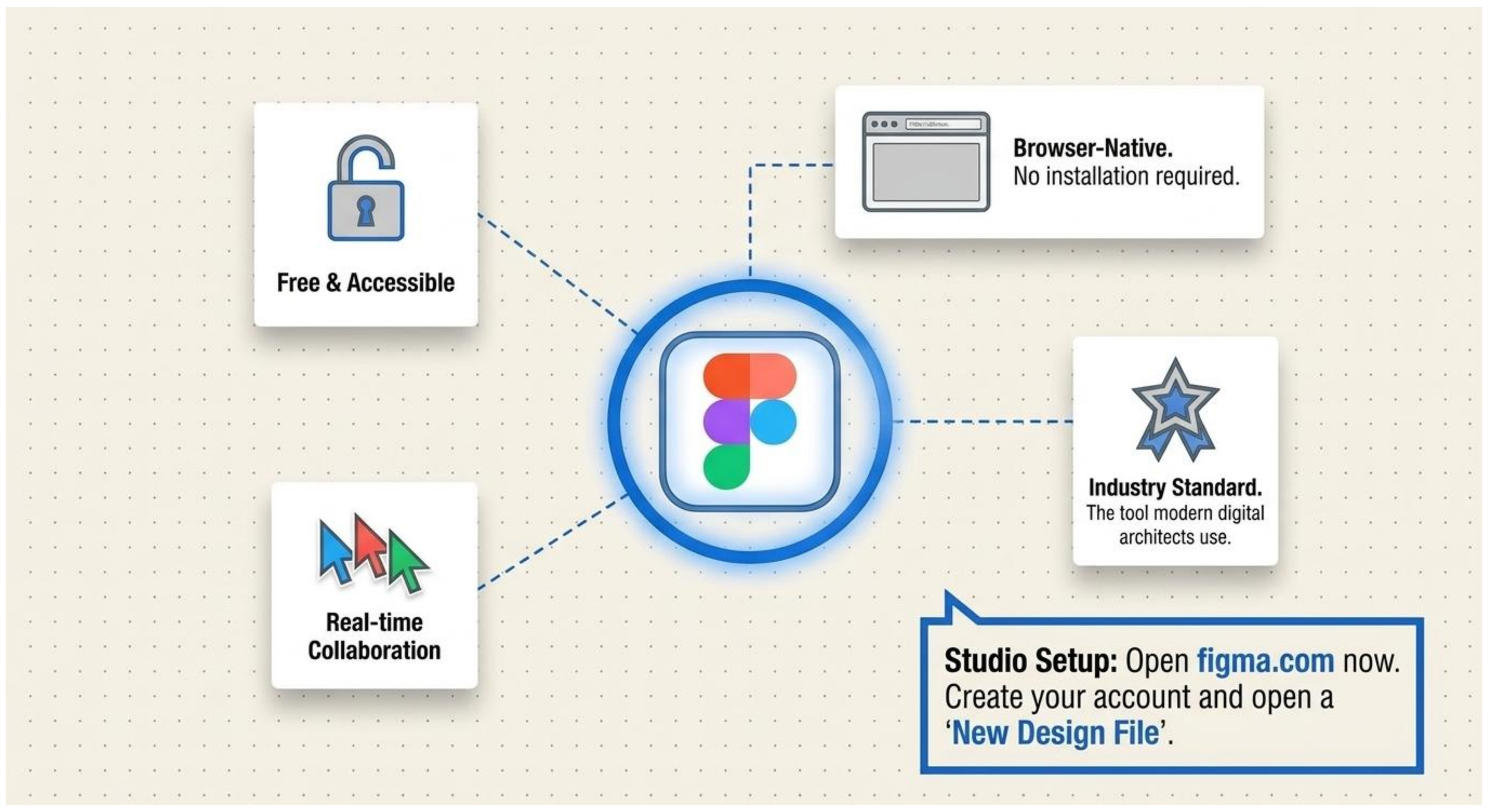
- **Focus:** Brand identity and visual polish.

PROTOTYPE

Attribute: Interaction

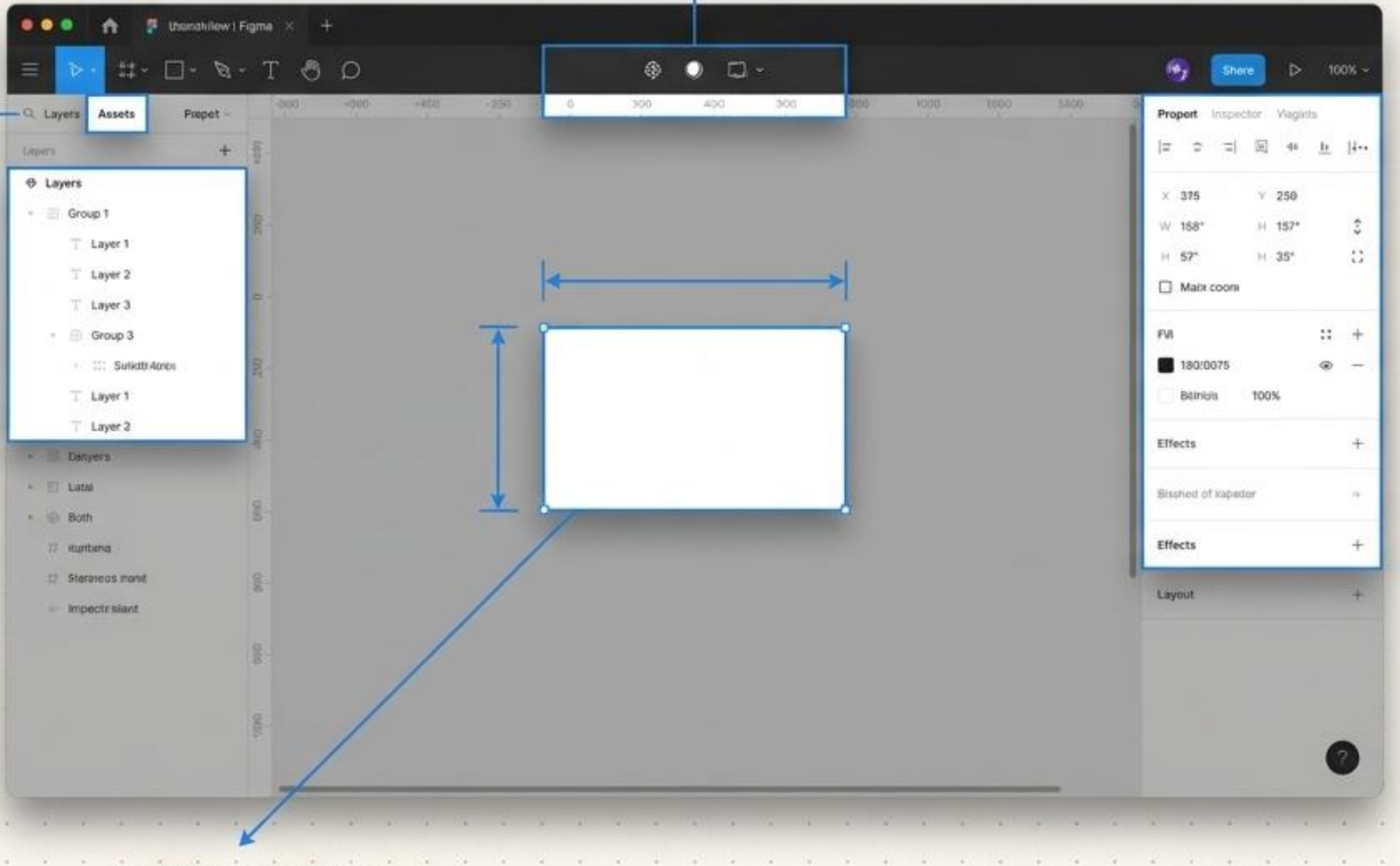


- **Focus:** User flow and clickable testing.


Free & Accessible
Browser-Native.
No installation required.
Real-time
Collaboration
Industry Standard.
The tool modern digital
architects use.
Studio Setup: Open figma.com now.
Create your account and open a
'New Design File'.

5. Assets: Your library of reusable components.

1. Toolbar: Where your fundamental building tools live.



3. Layers: The structural hierarchy of your elements.

4. Properties Panel: Controls for sizing, spacing, and styling.

2. Canvas: Your digital drafting board.

The Builder’s Toolkit

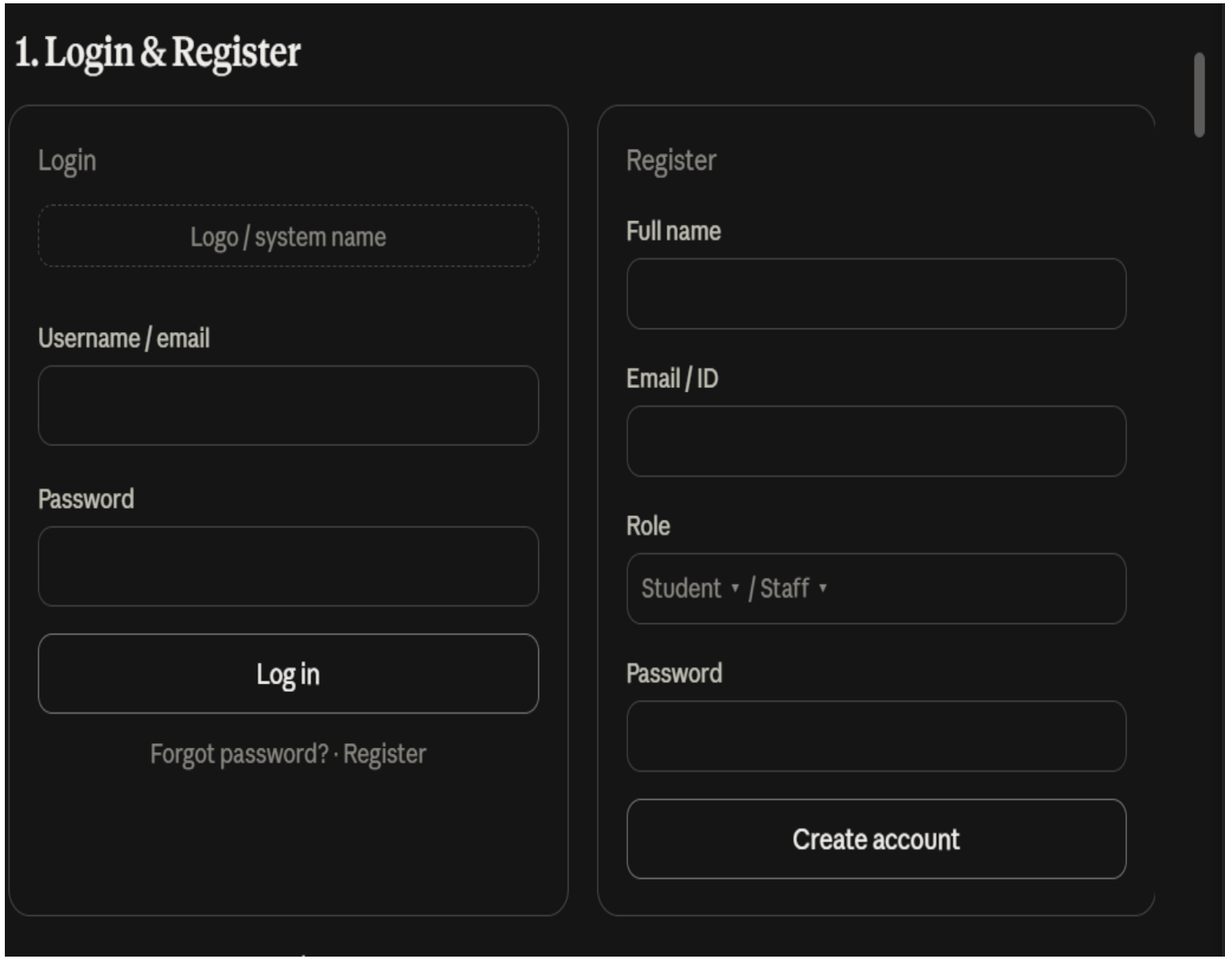
Speed is a byproduct of mastery. Put down the mouse and use your keyboard.

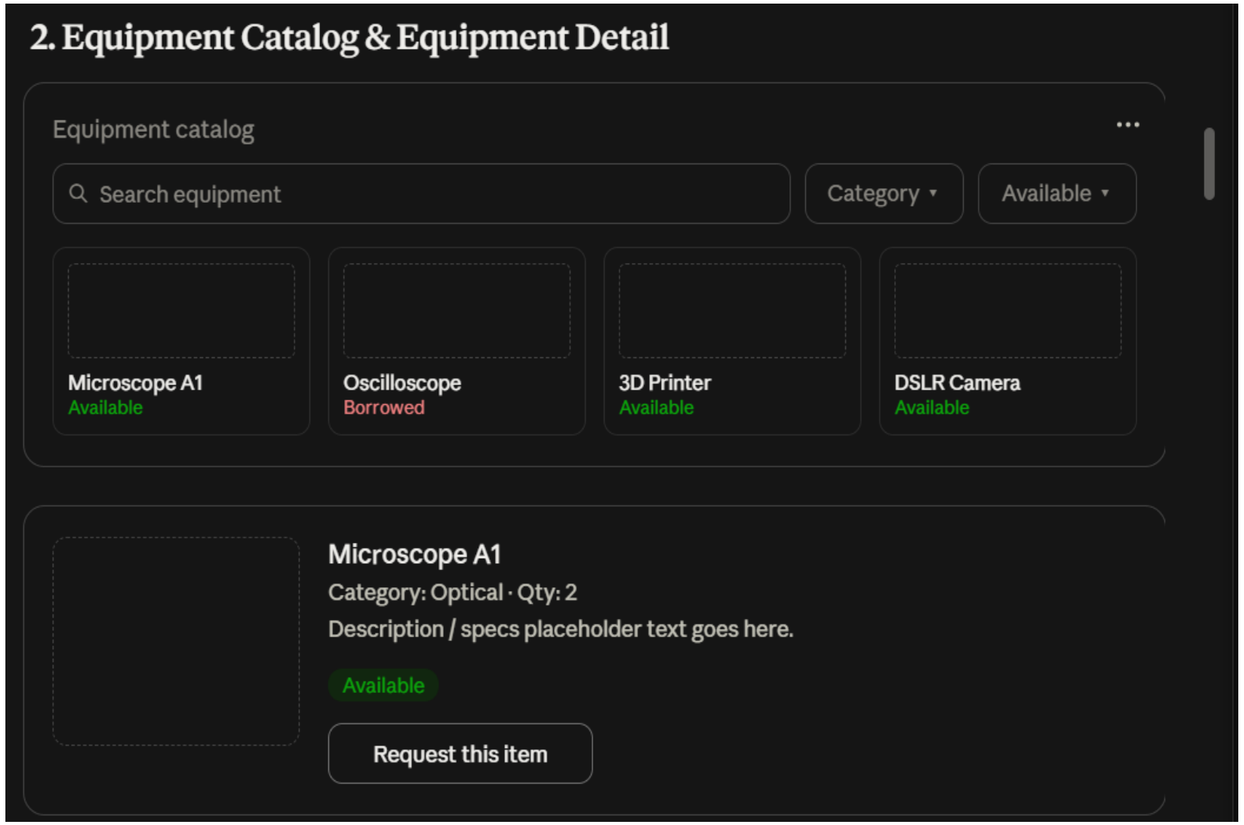




The **Frame** represents your application screen.
It is the **absolute boundary** of your blueprint.

The Complete Blueprint: Laboratory Equipment Reservation System





The Complete Blueprint: Laboratory Equipment Reservation System

3. Borrowing Request Form & My Requests

New borrowing request

Equipment

Microscope A1 (locked)

Borrow date

dd/mm/yyyy

Return date

dd/mm/yyyy

Research purpose

Submit request

My requests

Microscope A1 · 12–15 Sep	Pending
3D Printer · 1–3 Sep	Approved
DSLR Camera · 20–22 Aug	Returned
Oscilloscope · 5–6 Aug	Rejected

4. Staff Dashboard & Pending Requests

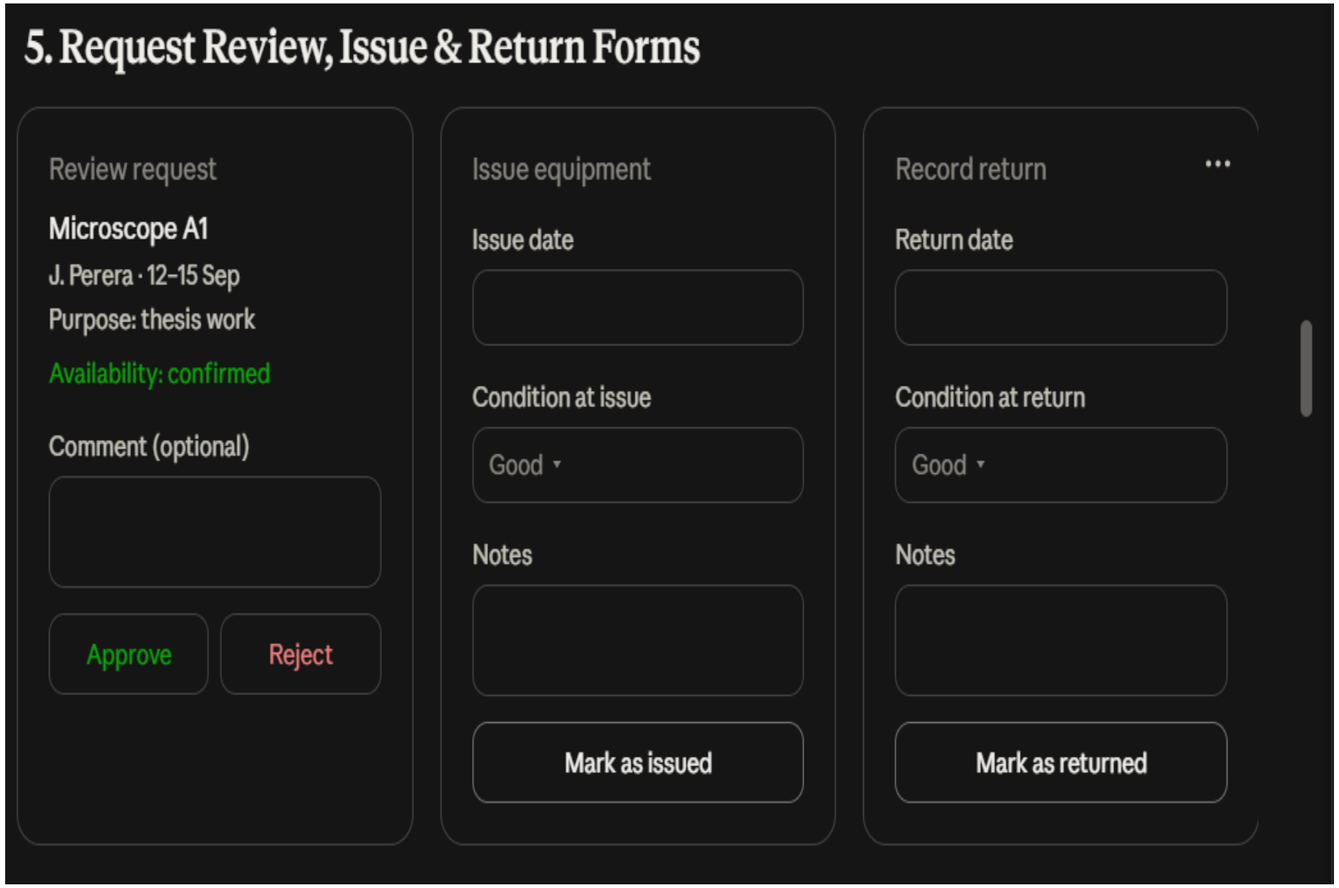
Pending requests	Due today	Overdue
7	3	1

Pending requests

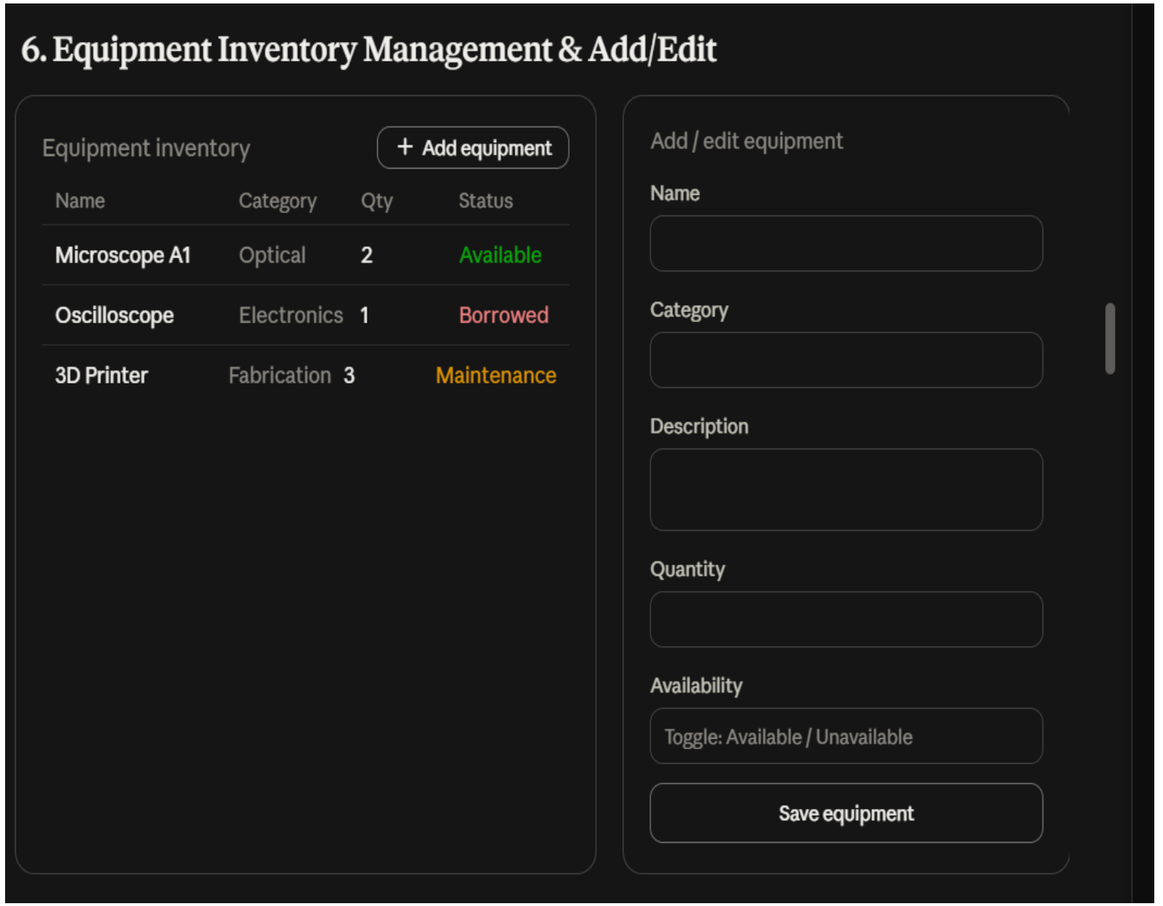
Requester / item	Dates	Purpose	Action
J. Perera · Microscope A1	12–15 Sep	Thesis work	Review
A. Silva · 3D Printer	14–16 Sep	Prototype	Review

The Complete Blueprint: Laboratory Equipment Reservation System

5. Request Review, Issue & Return Forms



6. Equipment Inventory Management & Add/Edit



This is an H1 heading

This is an H2 heading

This is an H3 heading

This is an H4 heading

This is an H5 heading

This is an H6 heading

B1 is for buttons

B2 is for buttons

L1 is for labels

L2 is for labels

P1 is for paragraphs: The quick brown fox jumps over the lazy dog.

P1 is for paragraphs: The quick brown fox jumps over the lazy dog.

P1 is for paragraphs: The quick brown fox jumps over the lazy dog.

P2 is for paragraphs: The quick brown fox jumps over the lazy dog.

P2 is for paragraphs: The quick brown fox jumps over the lazy dog.

P2 is for paragraphs: The quick brown fox jumps over the lazy dog.

P3 is for paragraphs: The quick brown fox jumps over the lazy dog.

P3 is for paragraphs: The quick brown fox jumps over the lazy dog.

P3 is for paragraphs: The quick brown fox jumps over the lazy dog.

P4 is for paragraphs: The quick brown fox jumps over the lazy dog.

P4 is for paragraphs: The quick brown fox jumps over the lazy dog.

P4 is for paragraphs: The quick brown fox jumps over the lazy dog.

in Gemini Notebook

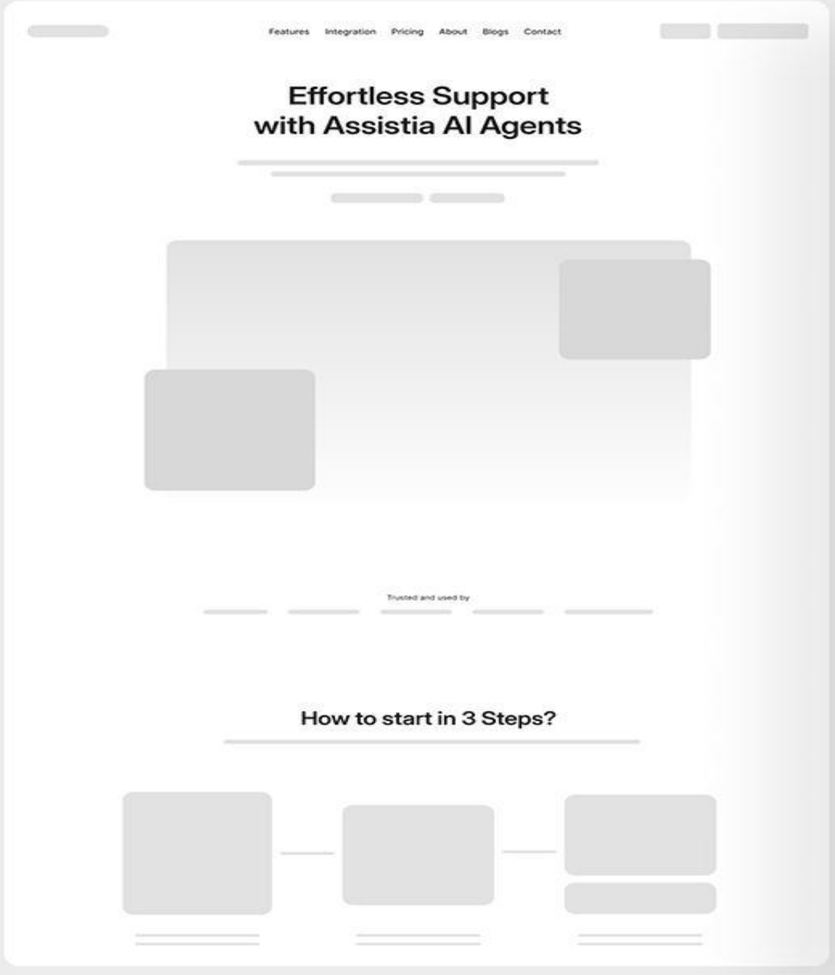
Wireframe (Structure)

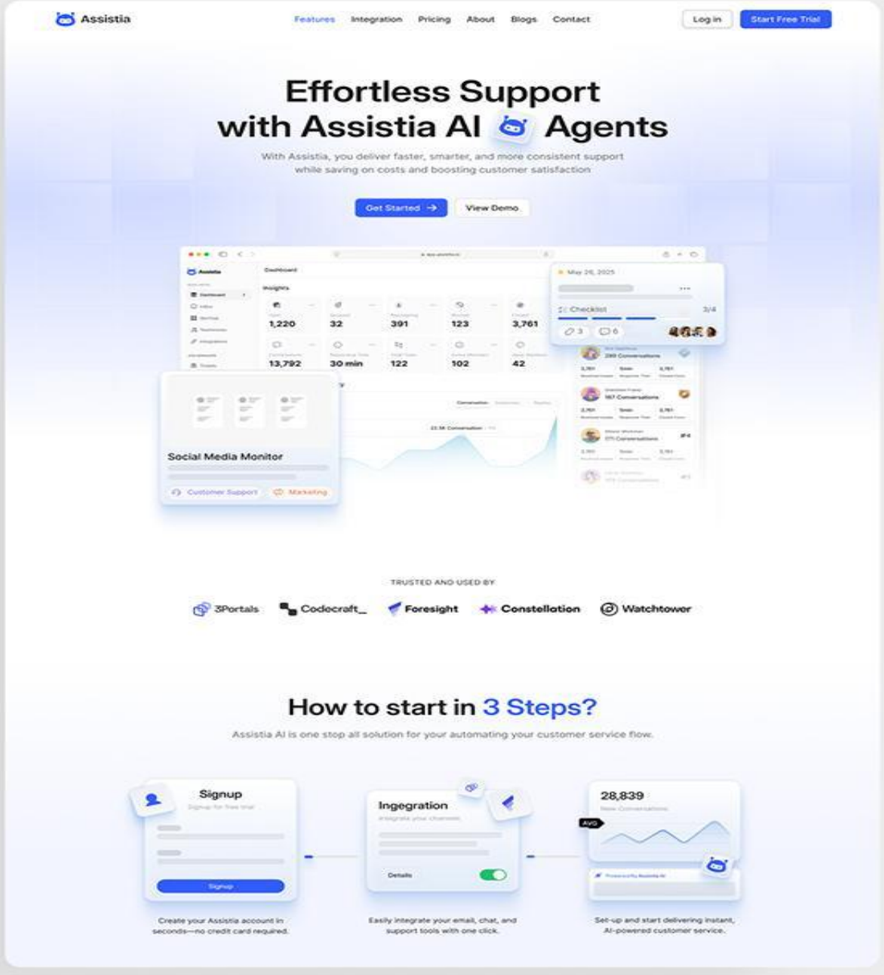
UI (Appearance)



Elevating the structure. Notice how the fundamental layout did not change—we only **applied colour, border radius, typography, icons**, and **refined spacing** on top.

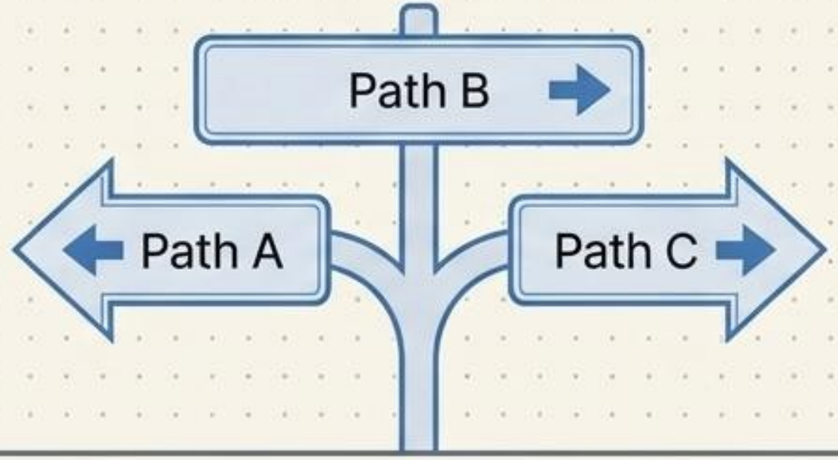
Wireframe → Visual Design





The Four Pillars of UX

1. Keep Navigation Obvious.



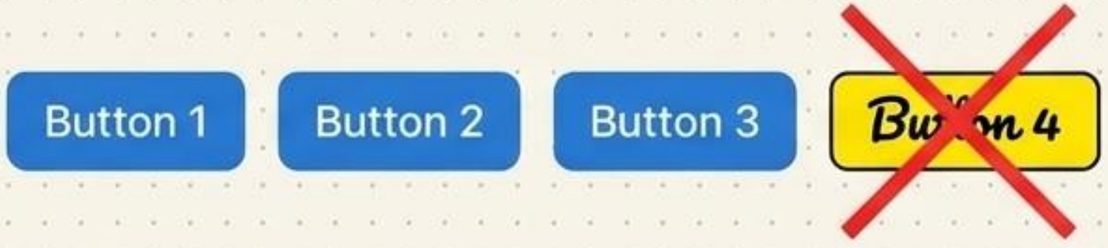
2. Give Feedback.



3. Prevent Errors.



4. Stay Consistent.



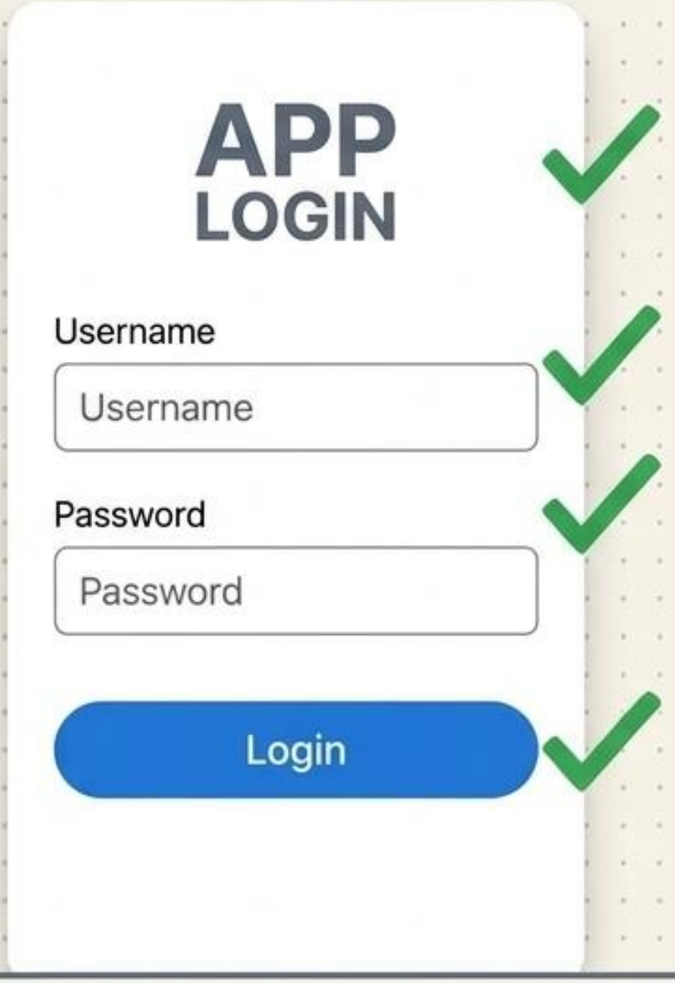
Gemini Notebook

Class Prompt: Train your eye. What specific UX and Grid rules did the left side violate?

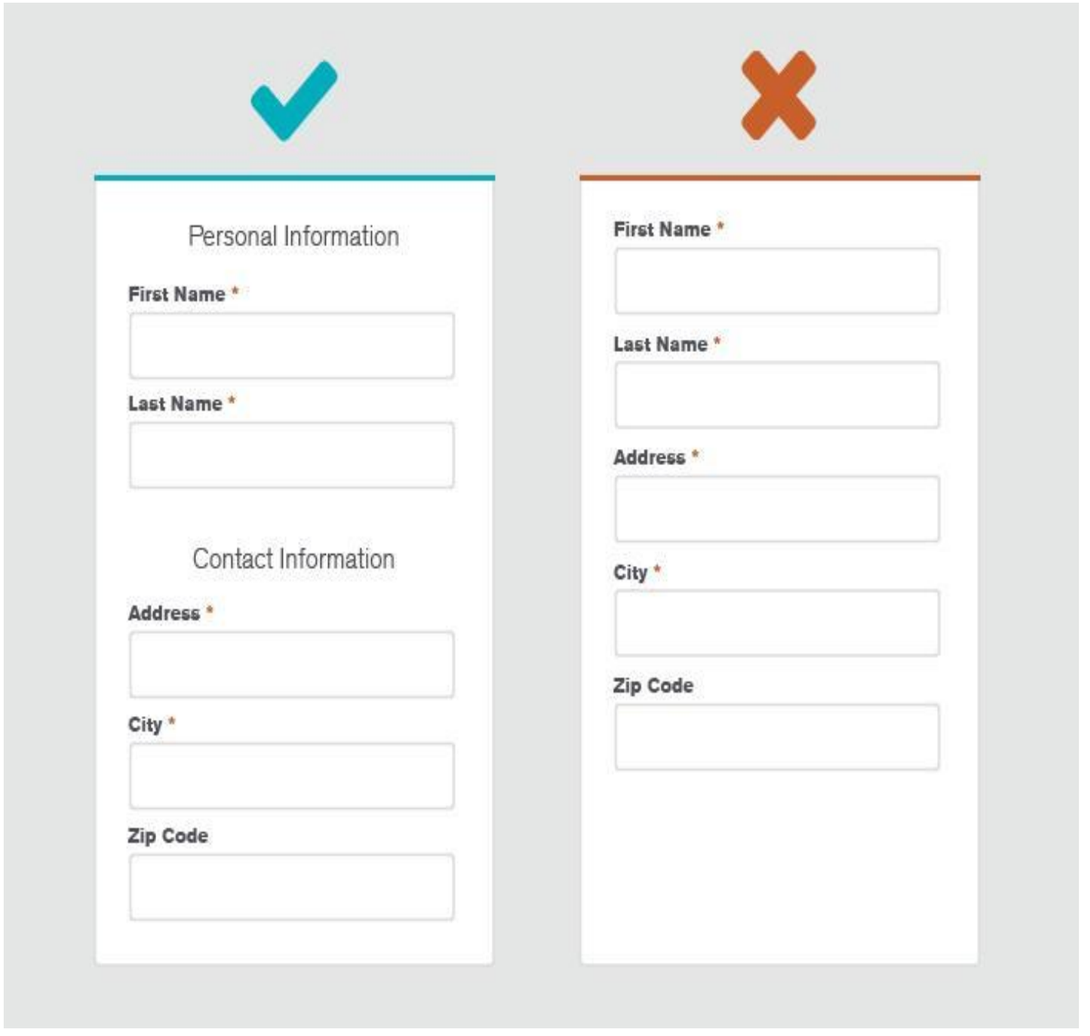
Bad UI



Good UI



Gemini Notebook



CLUTTERED: 6 FIELDS

Newsletter Signup

Email

you@example.com

Phone

555-123-4567 unnecessary

Fax

555-123-4568 unnecessary

Address

123 Main St unnecessary

City

Springfield unnecessary

Zip

62704 unnecessary

Subscribe

Too many fields for a simple newsletter signup.

STREAMLINED: 1 FIELD

Newsletter Signup

Email

you@example.com

Subscribe

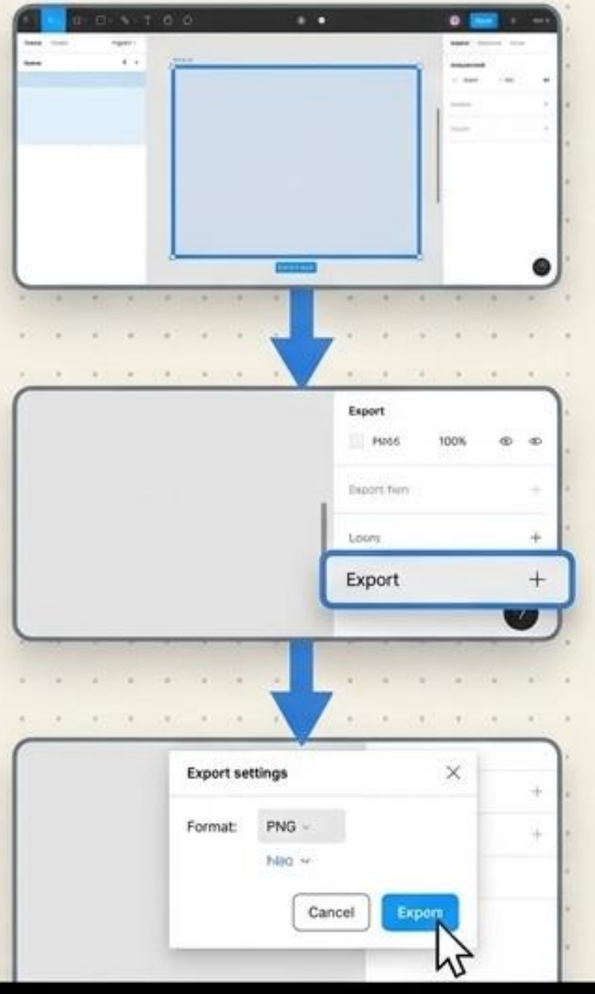
Only ask for what you actually need.

Task: Build Your First Screen

- [ ] Must include a prominent **Header** area.
- [ ] Must include **Main Navigation** layout.
- [ ] Must include a central **Form** area.
- [ ] Must include at least one clear **Action Button**.



Exporting Your Blueprint



1. Select the main **Frame** layer on the canvas.

2. Navigate to the **Export Panel** (bottom right of screen).

3. Set format to **PNG** and click **Export**.

Action: Save this PNG for your portfolio.

Homework: Design two additional screens for your system before next week's session.

Gemini Notebook